Armadale Kalamunda Group

Peri-Operative Management of Patients with Obstructive Sleep Apnoea Guideline

1. Purpose

To provide guidance for the peri-operative management of patients with Obstructive Sleep Apnoea (OSA) having surgery at Armadale Kalamunda Group (AKG).

2. Background

Patients with OSA (both diagnosed OSA, whether treated, partially treated, or untreated, and suspected OSA), undergoing procedures under anaesthesia are at an increased risk for perioperative complications when compared to those without this condition. Identifying patients with OSA and those at high risk for OSA before surgery using appropriate screening and diagnostic tools and implementing targeted perioperative precautions and interventions may help to reduce perioperative complications.

If available, consideration should be given to obtaining the results of the sleep studies and the patients should be instructed to bring their positive airway pressure (PAP) equipment to hospital for perioperative use if they use it routinely at home.

Additional evaluation to allow preoperative cardiopulmonary optimization should be considered in patients with diagnosed, partially treated/untreated, and suspected OSA where there is any indication of an associated significant or uncontrolled systemic disease, or additional problems with ventilation or gas exchange, such as hypoventilation syndromes, severe pulmonary hypertension, and resting hypoxemia in the absence of other cardiopulmonary disease.

3. Applicability

This guideline applies to nursing and medical staff within the peri-operative unit providing care for patients with OSA having surgery at AKG.

4. Guidelines

Current evidence indicates that the outcome and hence, the post-operative management requirements of OSA patients depend upon:

- Severity of OSA
- CPAP usage and compliance
- Type of surgery
- Post-operative analgesic requirements

Currently, there is inadequate evidence to recommend the use of routine sleep testing in the preoperative period in all patients suspected to have OSA. A balance needs to be struck between the desire to minimize postoperative complications and the responsible use of resources, to avoid creating undue economic, logistic and clinical burden on the health care system.

In our hospital, we shall follow an approach of risk stratifying patients according to STOP-BANG, which is the most widely validated scoring system for OSA and doing sleep studies based on the STOP-BANG score, the invasiveness of surgery and the requirement of pain relief in the post-operative period. This scoring system is described below in table 1.

Table 1: The STOP-BANG scoring system

No.	Question	Answer
1	Do you SNORE loudly?	Yes or No
2	Do you often feel tired, fatigued, or sleepy during the daytime?	Yes or No
3	Has anyone observed you stop breathing during your sleep?	Yes or No
4	Do you have or are you being treated for high blood pressure?	Yes or No
5	Are you obese/ very overweight – BMI more than 35 kg/m^2	Yes or No
6	Age over 50 years old?	Yes or No
7	Neck Circumference >40 cm (or 16 inches)?	Yes or No
8	Are you male?	Yes or No
Score: **If YES to 0 – 2, then low risk of sleep apnoea.** **If YES to 3 – 4 of the above, then you are at intermediate risk of having sleep apnoea.** **If YES to 5 – 8 of the above, then you are at high risk of having sleep apnoea**		

OSA is classified based on the results of sleep studies into mild, moderate and severe (Table 2). Apnoea-hypopnoea index (AHI) is defined as the number of apnoeic and/or hypopnoeic episodes per hour of sleep.

Table 2: Classification of OSA based on AHI

AHI	Severity of OSA
0-5	Normal
5-15	Mild OSA
15-30	Moderate OSA
>30	Severe OSA

5. Pre-operative management

- All patients for elective surgery with diagnosed or suspected OSA must be seen by an anaesthetist in pre-admission clinic (PAC) prior to surgery.
- All patients who are compliant with CPAP use at home will be instructed to bring in their machines on the day of surgery, barring which, their surgery would have a high likelihood of getting cancelled on the day.
- The PAC anaesthetist must use the STOP-BANG diagnostic questionnaire to identify the likelihood of OSA in suspected patients.

Sleep studies should be recommended for all patients having elective surgery with a STOP-BANG score equal to or more than 6 undergoing moderate risk procedures, and a score equal to or more than 3 undergoing high risk procedures. Exceptions: patients for category 1 surgery. If a patient is unable to have sleep studies or to use a CPAP machine, the anesthetist can use their discretion and proceed with surgery, treating them as “probable OSA”.

- All patients with BMI > 45, and symptoms or signs suggestive of pulmonary hypertension or right heart failure should be investigated and referred to tertiary centres if considered appropriate.
- The PAC anaesthetist should document a plan for the post-operative management of the patient, notify the float nurse about these and discuss with the Intensive Care Unit consultant if considered necessary.
- OSA patients who would require oximetry on the ward, HIVE or ICU admission post-operatively should have a handover (OneNote) completed by the PAC or float nurse so that they are flagged on the Theatre Management System.
- The decision to accept a patient with OSA as a day case or not will also be determined by optimization of the comorbid conditions (such as hypertension, arrhythmias, heart failure, cardiovascular disease and metabolic syndrome), and the ability to use CPAP after discharge.

Table 3: Management guideline for patients with known or suspected OSA

Categories of patients	**Low Risk Surgery** *	**Intermediate Risk Surgery** *	**High Risk Surgery** *
1. Low risk of OSA (STOP-BANG < or = 3) 2. OSA managed with CPAP	Normal ward bed OK for discharge home if suitable	Normal ward bed OK for discharge home if appropriate	Ward bed with oximetry, HIVE bed or ICU
1. High risk of OSA (STOP-BANG > 3) 2. OSA not on CPAP	Ward bed with oximetry OK for discharge home if suitable	Ward bed with oximetry, HIVE bed or ICU	HIVE bed or ICU

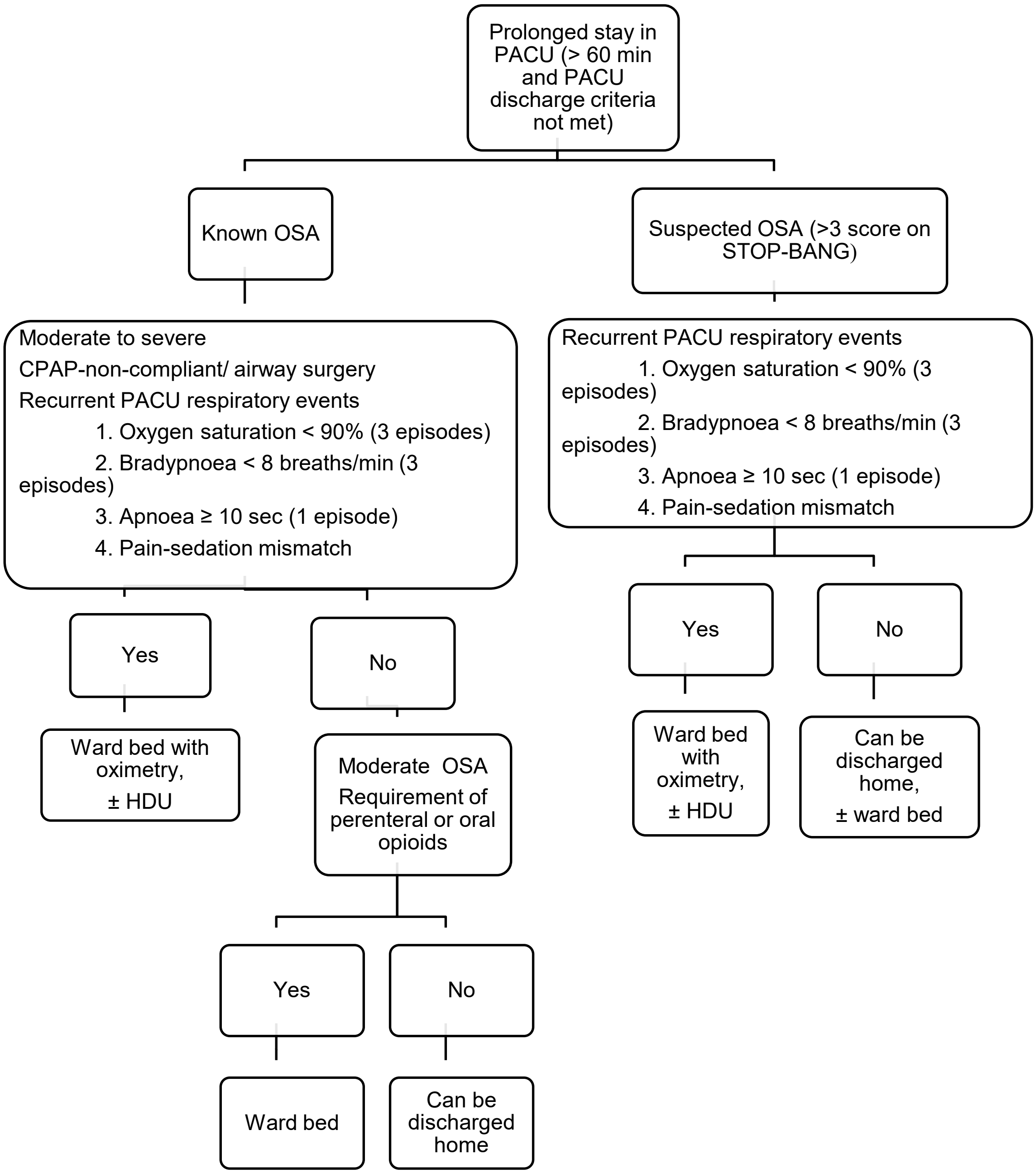
* The risk of surgery here is just on the basis of opioid use and post-operative respiratory complications, see table 4.

Table 4: Classification of surgeries based on risk of post-operative respiratory complications and opioid use

Classification of surgical procedures based on risk of post-operative respiratory complications and opioid use:	
Low risk	All peripheral and small vascular surgeries, gastro-intestinal endoscopic procedures, arthroscopies, all surgeries under regional anaesthesia without neuraxial morphine.
Intermediate risk	Laparoscopic abdominal surgeries, total abdominal hysterectomies.
High risk	Laparotomies, joint replacements, large ventral hernia repairs, upper airway surgeries (not able to use CPAP), UPPP, big tonsils causing OSA, all surgeries requiring post-op PCA or having high post-operative opioid requirements (significantly higher than their baseline), all patients receiving neuraxial morphine.

Figure 1: Flowchart for post-operative management of patients who have prolonged recovery time or post-operative concerns in PACU (adapted for AHS from the Canadian Anaesthesiologists' Society guidelines)

Note- Pain-sedation mismatch implies that a patient has simultaneous occurrence of high pain scores and high sedation levels.



6. Post-operative management

- Most OSA patients do not need HDU or ICU placement unless there are other co-existing medical or surgical reasons.

- Ward bed with oximetry requires that a patient is admitted in an area with an adequate nurse to patient ratio, continuous pulse oximetry with appropriately set alarms, and adequate nursing training and experience in the use of CPAP.
- Patients monitored remotely via HIVE should have continuous pulse oximetry with appropriately set alarms, and a nurse experienced in the use of CPAP.
- All patients who use CPAP at home should start using their machines as soon as possible after surgery, preferably in the post-anaesthesia care unit.
- The target oxygen saturation should be 92-95% to minimise depression of respiratory drive and consequent hypoventilation.

7. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) Consent to Treatment Policy OD 0657/16
- HDWA Clinical Handover Policy MP 0095
- HDWA Recognising and Responding to Acute Deterioration Policy MP 0086/18
- East Metropolitan Health Service (EMHS) Patient Identification and Procedure Matching EMHS: 21

8. Supporting information

The following documents support the implementation of this policy:

- Post-Procedure Care in the Same Day Unit THE-PRO-0505-19
- Preoperative Screening Investigations Procedure THE-PRO-0374-10
- Recognition and Response to Acute Deterioration AKG-POL-0388-14
- Patient Controlled Intravenous Analgesia ANA-POL-0040-11
- EMHS Consent Policy EMHS: 5

9. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Specific findings are to be reported at the departmental meeting for review and identification of any opportunities for improvement.

10. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 – Action 5.3 Partnering with consumers
- Clinicians use organisational processes from the Partnering with Consumers Standard when providing comprehensive care to:
    - a. Actively involve patients in their own care
    - b. Meet the patient’s information needs
    - c. Share decision-making

- Standard 5 – Action 5.10 Screening of risk
- Clinicians use relevant screening processes:

  a. On presentation, during clinical examination and history taking, and when required during care

  b. To identify cognitive, behavioural, mental and physical conditions, issues, and risks of harm

  c. To identify social and other circumstances that may compound these risks

- Standard 8 – Action 8.4 Recognising acute deterioration
- The health service organisation has processes for clinicians to detect acute physiological deterioration that require clinicians to:
    - a. Document individualised vital sign monitoring plans
    - b. Monitor patients as required by their individualised monitoring plan
    - c. Graphically document and track changes in agreed observations to detect acute deterioration over time, as appropriate for the patient

Standards Australia/New Zealand Standards (AS/NZS):

- ANZCA PS 15; 2016: Guidelines for the Perioperative Care of Patients Selected for Day Stay Procedures
- ANZCA PS 15; BP 2016: Recommendations for the Perioperative Care of Patient Selected for Day Stay Procedures

11. References

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12. Glossary

The following definitions are relevant to this policy.

Term	Definition
Obstructive sleep apnoea	A disorder in which breathing is repeatedly interrupted during sleep due to repetitive collapse of the upper airway.
Apnoea	Complete cessation of breathing that lasts 10 seconds or more.
Hypopnoea	50% reduction in airflow associated with desaturation and/ or arousal from sleep.
Apnoea-hypopnoea index (AHI)	The number of apnoeic and/or hypopnoeic episodes per hour of sleep.

13. Document owner

Head of Department Anaesthetics

Enquiries relating to this guideline may be directed to:

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14. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
3.0	25/06/2025	25/06/2028	Table 3 – Significant changes Table 4 – Intrathecal has been changed to Neuraxial

15. Authorisation

This guideline has been authorised and issued by the Director of Clinical Services.

Authorised by	Dr Tania Strickland – Director of Clinical Services
Authorisation date	25/06/2025
Published date	25/06/2025